

# Chapter 6 - The Sine-Gordon Equation - Notes

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## 1 The Sine Gordon Equation

Sine Gordon is good for wave fronts.

$$u_{tt} - u_{xx} = \sin(u) = 0$$

We will look at solutions to the traveling wave equation using the ansatz below:

$$u(x, t) = f(x - ct)$$

Let's take some derivatives:

$$\begin{aligned} u_t &= -cf'(x - ct) & u_{tt} &= c^2 f''(x - ct) \\ u_x &= f'(x - ct) & u_{xx} &= f''(x - ct) \end{aligned}$$

Now we can substitute into  $u_{tt} - u_{xx} + \sin(u)$ :

$$c^2 f'' - f'' + \sin(f) = 0$$

We can form a nicer equation by multiplying by  $f'$

$$\begin{aligned} f'(c^2 f'' - f'' + \sin(f)) &= 0 \\ c^2 f'' f' - f'' f' + \sin(f) f' &= 0 \\ f'' f' (c^2 - 1) + \sin(f) f' &= 0 \end{aligned}$$

We can now integrate to produce a first order equation

$$\frac{1}{2}(c^2 - 1)(f')^2 - \cos(f) = a$$

with  $a$  as some constant of integration.  $a$  can be different depending on the problem. To determine  $a$ , we apply the decay conditions for a traveling wave front. As the wave extends toward infinity ( $y \rightarrow \infty$ ), the wave profile must flatten out. Therefore, we assume the decay conditions:

$$\lim_{y \rightarrow \infty} f(y) = 0 \quad \text{and} \quad \lim_{y \rightarrow \infty} f'(y) = 0$$

Taking the limit of our integrated equation as  $y \rightarrow \infty$ :

$$\begin{aligned} a &= \lim_{y \rightarrow \infty} \left[ \frac{1}{2}(c^2 - 1)(f'(y))^2 - \cos(f(y)) \right] \\ a &= 0 - \cos(0) \\ a &= -1 \end{aligned}$$

Substituting  $a = -1$  back into our equation allows us to separate variables and solve for the wave front (kink) solution.

Substituting  $a = -1$  back into our first-order equation gives:

$$\frac{1}{2}(c^2 - 1)(f')^2 - \cos(f) = -1$$

We can rearrange this equation to solve for  $(f')^2$ :

$$(f')^2 = 2 \left( \frac{1 - \cos(f)}{1 - c^2} \right)$$

Taking the square root of both sides gives us a separable differential equation:

$$f' = \pm \sqrt{\frac{2}{1 - c^2}} \sqrt{1 - \cos(f)}$$

Now, we separate the variables so that all terms with  $f$  are on one side and terms with  $x$  are on the other:

$$\frac{df}{\sqrt{1 - \cos(f)}} = \pm \sqrt{\frac{2}{1 - c^2}} dx$$

You are smart and can figure out the rest using partial fractions and stuff.